

PROJECT : INSURANCE CHARGE PREDICTOR

Problem Statement:

The objective is to build a model that predicts the insurance charges for a customer based on the various parameter. The input & output is in the format of Numbers and The requirement is clear and the input & output for previous records is also already given. So we are planning to go with the ***Machine Learning - Supervised Learning - Regression.***

About Dataset:

The given dataset contains **5 input** columns(**age,sex,bmi,children,smoker**) and **1 output** column (**charges**) - This is estimated based on the requirement and the output is determined based on what we need to predict.

There are totally 1338 records present in the table. It has 1338 records and 6 columns.

Preprocessing Data:

Out of 5 input columns - **Sex and Smoker is a Categorical Column** and both are **ordinal data**. So we have to do **Mapping Label Encoder** on this columns which will result in no column extension but equivalent 0s and 1s to the respective column.

Model	R2_score
Multiple Linear Regression	~75%
Support Vector Machine	~ -79% Tried changing the kernel and added penalty too , SVM(R) doesnt support this data good
Decision Tree	~87% (with pruning)
Random Forest	~85%

Decision Tree has better result among all the other Regression Model that were used , so the model is deployed with Decision Tree Algorithm as "Insurance_charge_predictor.sav".

The Tried Models and its code is present in the GitHub repository :

https://github.com/KanisgarPrabu/1.Python/tree/main/Python_Assignment/Week3_ML_Regression/Regression_Final_Assignment